

What is AWS Load Balancer?

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AWS Load Balancer is a service that automatically distributes incoming network traffic across multiple servers (like EC2 instances) to make applications more reliable, fast, and highly available.

It helps to:

- Avoid overload on one server
- Redirect traffic to healthy servers
- Works with Auto Scaling to handle sudden traffic spikes.

Types of AWS Load Balancers:

1. Application Load Balancer (ALB) :

- Works at Layer 7 (HTTP/HTTPS)
- Best for web apps, path-based routing

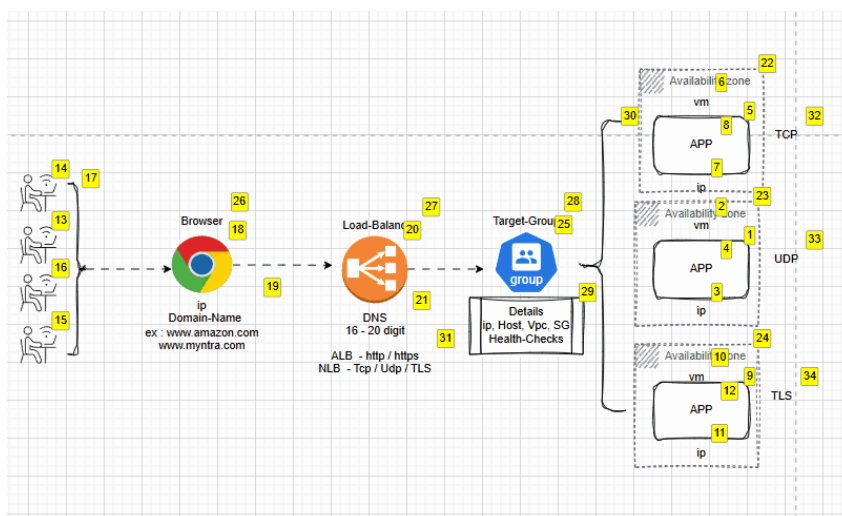
2. Network Load Balancer (NLB)

- Works at Layer 4 (TCP/UDP)
- Best for high performance, low latency

3. Gateway Load Balancer (GWLB)

- Used for third-party virtual appliances (firewalls, etc.)

Architecture of AWS Load Balancer :



Let's walk through the practical steps to set up an AWS Load Balancer with EC2 servers

- **EC2 Dashboard > Launch Instance**
- **Amazon Linux 2** or **Ubuntu**
- Choose **t2.micro** (Free Tier)
- Add **HTTP (port 80) & SSH (port 22)** in security group
- Launch at least **2 instances** in **different Availability Zones**.

Step 2 : Connect to EC2 Instances and Download the Packages(Like httpd) in both servers.

```

~\##### Amazon Linux 2
~\#####
~\###| AL2 End of Life is 2026-06-30.
~\#/
~\V~'-'->
~~~~
~~~~
~..-
~\/-\
~\m/'-\

```

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
<https://aws.amazon.com/linux/amazon-linux-2023/>

```

[ec2-user@ip-172-31-93-124 ~]$ sudo su -
[root@ip-172-31-93-124 ~]# yum install httpd

```

- **Systemctl start httpd**

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- **Systemctl enable httpd**
- **Systemctl status httpd** (To the httpd server is active or not.!))

```
[root@ip-172-31-93-124 ~]# systemctl start httpd
[root@ip-172-31-93-124 ~]# systemctl enable httpd
Created symlink from /etc/systemd/system/multi-user.target.wants/httpd.service to /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-93-124 ~]#
[root@ip-172-31-93-124 ~]#
[root@ip-172-31-93-124 ~]# vi /var/www/html/index.html
[root@ip-172-31-93-124 ~]#
```

Now, Create the web page using the vi command :

The httpd webserver default path is store web files **/var/www/html** , so we need to create a file in this path.

Command to Create a file : **vi index.html/var/www/html**

- Now **vi** will open a file and place HTML code into it and save the file .

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Arun Kumar Akula - DevOps Engineer</title>
  <style>
    body {
      font-family: 'Segoe UI', sans-serif;
      background: linear-gradient(135deg, #ff416c, #9d50bb, #1e90ff);
      background-size: 400% 400%;
      animation: gradientBG 15s ease infinite;
      color: #fff;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }

    @keyframes gradientBG {
      0% { background-position: 0% 50%; }
      50% { background-position: 100% 50%; }
      100% { background-position: 0% 50%; }
    }

    .card {
      background: rgba(255, 255, 255, 0.1);
      border-radius: 20px;
      padding: 30px 40px;
      text-align: center;
      box-shadow: 0 8px 20px rgba(0, 0, 0, 0.3);
      backdrop-filter: blur(10px);
      max-width: 400px;
    }

    h1 {
      font-size: 34px;
      background: linear-gradient(to right, #ffe53b, #ff2525);
      -webkit-background-clip: text;
      -webkit-text-fill-color: transparent;
      margin-bottom: 12px;
    }

    h2 {
      font-size: 20px;
      color: #ffeb3b;
      margin-top: 0;
    }
  </style>
</head>
<body>
  <h1>Arun Kumar Akula</h1>
  <h2>DevOps Engineer</h2>
</body>
</html>
```

In Server-1 vi file, I mentioned the code as '**Arun Kumar Akula**', and in Server-2 vi file, I used '**Arun Kumar**'. This was done to observe traffic distribution when we refreshed the page."

AWS Placement Groups?

Placement Groups in AWS are used to control how EC2 instances are placed within the AWS infrastructure to improve performance, availability.

Why Attach Placement Groups to EC2 Instances?

You use Placement Groups to get better performance, reliability, or fault isolation depending on your app needs.

Types of Placement Groups :

1. **Cluster:** Places instances close together for high speed and low latency.
2. **Spread:** Places instances on separate hardware to reduce failure risk.
3. **Partition:** Groups instances across separate racks for fault tolerance.

Step 4 : Create the placement groups

- Go to EC2 Dashboard in the AWS Console.
- In the left menu, click Placement Groups.
- Click Create placement group.
- Give it a name (e.g., my-placement-group).
- Choose the strategy: **Cluster, Spread, or Partition**
- Click Create placement group.

The screenshot shows the AWS Management Console interface for creating a new placement group. The top navigation bar includes the AWS logo, a search bar, and user information (United States (N. Virginia), Arunaja). The breadcrumb trail indicates the path: EC2 > Placement groups > Create placement group. The main heading is 'Create placement group' with an 'Info' link. Below this is a 'Placement group settings' section with three fields: 'Name' (containing 'Aws LB-placement groups'), 'Placement strategy' (a dropdown menu set to 'Cluster'), and 'Tags - optional' (a section for adding tags, with a note that no tags are currently associated and a button to 'Add new tag'). At the bottom right of the settings section are 'Cancel' and 'Create group' buttons. The footer contains 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

aws [Alt+S] United States (N. Virginia) Arunaja

EC2 > Placement groups > Create placement group

Create placement group [Info](#)

Placement group settings

Name

Placement strategy
Determines how the instances are placed on the underlying hardware.

Cluster

Tags - optional
No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Create group](#)

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Placement groups (1)						Actions	Create placement group
Find Placement Group by attribute or tag						< 1 >	
Group name	Group Id	Strategy	State	Partition			
☐ Aws LB-placement groups	pg-091a03d42a98196d7	cluster	available	-			

What is a Target Group in AWS :

A Target Group is a collection of servers (like EC2s) that a Load Balancer sends traffic to, and it also stores all the details related to those servers, **like IPs, ports, health check status, and protocols.**

Why does a Load Balancer attach to a Target Group:

The Load Balancer doesn't talk to EC2s directly it talks to **Target Groups**, which manage the actual servers.

Step 5 : Create a Target Group

- Go to the EC2 Dashboard in AWS Console.
- On the left menu, click Target Groups
- Click Create target group

Fill the Required Details:

- Target type: Choose Instances (for EC2)
- Target group name: Example — my-target-group
- Protocol: HTTP
- Port: 80
- VPC: Choose the same VPC as your EC2 instances
- Protocol version: HTTP1

Configure Health Checks:

- Protocol: HTTP
- Path: / (default)

Click Next to register targets

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- Select the EC2 instances you want to include
- Click **Include as pending**

Register targets
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Instance ID	Name	State	Security groups
i-0d231b886a5dc34c9	Aws Load-Balancer	Running	launch-wizard-23
i-0d5f94b8db05d6132	Aws Load-Balancer	Running	launch-wizard-22

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.
80
1-65535 (separate multiple ports with commas)

Include as pending below

- Click **Create target group**

Review targets

Targets (2)

Remove all pending

Filter targets

Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-0d5f94b8db05d6132	Aws Load-Balancer	80	Running	launch-wizard-22	us-east-1d	172.31.29.78	subnet-00fba79f2b
i-0d231b886a5dc34c9	Aws Load-Balancer	80	Running	launch-wizard-23	us-east-1c	172.31.93.124	subnet-024f5f040b

2 pending

Cancel Previous **Create target group**

- Now we can see the created Target group in the Target groups Dashboard.

Target groups (1) Info

Filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer
AwsLB-Target	arn:aws:elasticloadbalancin...	80	HTTP	Instance	None associated

Step 6 : Create an Application Load Balancer (ALB)

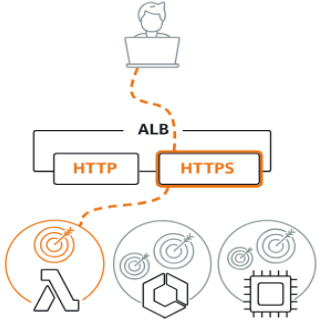
- Go to the EC2 Dashboard
- In the left menu, click Load Balancers
- Click Create Load Balancer

- Choose Application Load Balancer

[EC2](#) > [Load balancers](#) > Compare and select load balancer type

Load balancer types

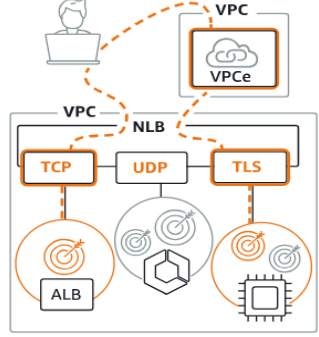
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

► **Classic Load Balancer - previous generation**

Fill the Basic Configuration :

- Name: e.g., my-alb
- Scheme: Internet-facing (for public access)
- IP address type: IPv4
- VPC : Choose the same VPC as your EC2 instances
- Availability Zones and subnets : Select at least 2 Availability Zones — the same ones where your EC2 instances are running
- Security groups : Choose the same Security Group that your EC2 instances are using.
- Listeners: Keep HTTP on port 80
- Click **Create Load Balancer** (After a few minutes, your ALB will be ready.)

Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

[Filter load balancers](#)

☐	Name	DNS name	State	VPC ID	Availability Zones	Type	Date create
☐	ALB	ALB-225366186.us-east-1.e...	Active	vpc-0b9c82995f074d71b	2 Availability Zones	application	June 25, 20...

- Go to Load Balancer > Description Tab
- Copy the DNS name

MY-LB

Actions ▾

▼ Details

Load balancer type

Application

Status

Active

VPC

[vpc-0b9c82995f074d71b](#)

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z35SXDOTRQ7X7K

Availability Zones

[subnet-024f5f040b92bbf49](#) us-east-1c (use1-az2)
 [subnet-00fba79f2bacd836d](#) us-east-1d (use1-az4)

Date created

June 25, 2025, 21:37 (UTC+05:30)

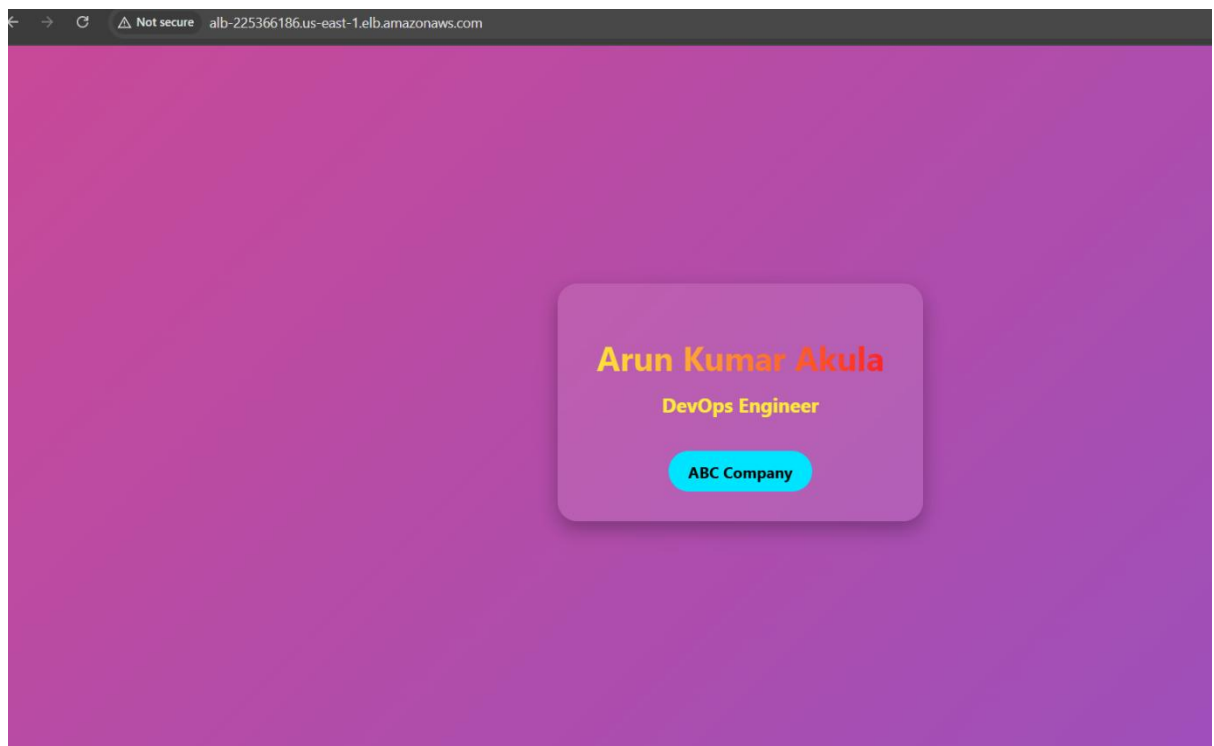
Load balancer ARN

[arn:aws:elasticloadbalancing:us-east-1:869546918261:loadbalancer/app/MY-LB/825b46bb2d4d5c34](#)

DNS name Info

[MY-LB-1884880434.us-east-1.elb.amazonaws.com](#) (A Record)

- Paste in your browser to test it.!



Load Balancer Output :

- **Server 1 shows:** Arun Kumar Akula
- **Server 2 shows:** Arun Kumar

When you open the Load Balancer DNS and refresh the page, the output changes between both this shows traffic is going to both servers correctly.

